BUSTER: Adaptive Sampling for VLM-guided Unsupervised Video Anomaly Detection

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Abstract

Vision-Language Models have shown strong promise for video anomaly detection, but applying them to long surveillance streams remains computationally impractical. Sparse uniform sampling further suffers from temporal aliasing and can miss short-lived anomalies, while many existing approaches also depend on expensive human supervision. We address these limitations with BUSTER, a label-free framework that distills VLM knowledge into a lightweight streaming detector for adaptive video anomaly detection. BUSTER follows a Scout-and-Burst policy: a lightweight student monitors the stream in Scout mode at low frame rate and selectively switches to Burst mode for dense inspection only around anomaly candidates. To make this possible without manual labels, we mine semantic pseudo-labels from a VLM teacher and train the student with a rank-aware distillation objective that preserves relative anomaly severity rather than only matching absolute scores. This design enables VLM-guided behavior during continuous monitoring without incurring VLM-level cost on the full stream. Experiments on UCF-Crime, XD-Violence and ShanghaiTech show that BUSTER achieves state-of-the-art performance among unsupervised methods, matches dense fixed-rate processing while using only 53.2 percent of frames, its detection path runs at 496.7 FPS corresponding to 16.6x real time at 30 FPS, and supports on-demand segment-level explanations through burst-triggered VLM reasoning. These results show that adaptive distillation can make label-free VLM-guided anomaly detection both efficient and practical for real-world surveillance.

1 Introduction

Vision-Language Models (VLMs) have recently boosted a wide range of downstream tasks by injecting rich semantic priors and strong reasoning capabilities. Video Anomaly Detec-

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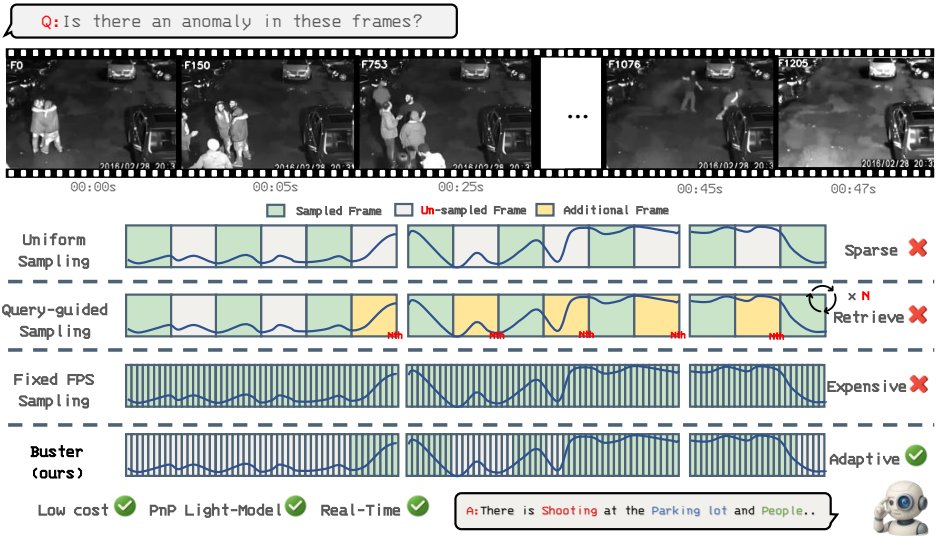


Figure 1: Illustration of sampled and unsampled frames under different policies. Compared to uniform, query-guided, and fixed-rate sampling, BUSTER adaptively allocates additional frames only around anomaly candidates and processes burst segments for detection and explanation.

tion (VAD) is likewise evolving beyond score-based detection toward more *semantic* and *interpretable* understanding, making VLM-guided VAD an increasingly compelling direction [1, 16, 26]. Despite this promise, naively applying VLM reasoning to long, untrimmed surveillance videos remains impractical at scale, and existing frame sampling strategies expose a clear trade-off, as illustrated in Fig. 1, uniform sparse sampling risks skipping short-lived anomalous cues, query-driven sampling can recover missing evidence via iterative retrieval but adds latency and system complexity that are undesirable for continuous monitoring, and fixed-rate dense processing improves coverage but incurs substantial overhead [22, 25, 27].

However, *adaptive sampling* itself remains under-explored in VLM-guided VAD. Existing attempts [34] improve efficiency by reusing a lightweight VAD model as an anomaly scorer and then redistributing frame density according to its scores. While effective for anomaly-focused selection, this still treats temporal budget allocation as a byproduct of generic anomaly scoring, rather than learning a lightweight controller for the sampling objective itself. For long surveillance streams, however, the key challenge is not only to identify suspicious moments, but to decide when additional temporal evidence should be acquired under a limited compute budget. This motivates learning a lightweight sampling controller that makes content-aware switching decisions while inheriting semantic guidance from a VLM teacher.

BUSTER first uses a teacher VLM to derive binary anomaly confidence scores from Yes/No token normalization over temporal windows, then distills this signal into a lightweight recurrent student equipped with temporal-difference features. Rather than treating the student as a detector only, we use it as an online sampling policy: normal content is processed sparsely in *Scout* mode, whereas high-score intervals are revisited at higher temporal resolu-

tion in *Burst* mode. This turns adaptive sampling into a learned, task-specific component instead of a fixed heuristic or an external off-the-shelf selector. As a result, BUSTER preserves VLM-informed anomaly sensitivity where needed, while reserving expensive reasoning for short burst segments that can also be used for interpretable explanation. Extensive experiments show that this design yields a favorable efficiency-accuracy trade-off in practice. On UCF-Crime, BUSTER matches dense fixed-rate processing while using only 53.2% of the frames, achieves state-of-the-art performance among unsupervised methods, and maintains robust generalization under cross-dataset evaluation. This design directly addresses the long-stream challenge above by turning temporal budget allocation into a learned, content-aware decision rather than a byproduct of generic anomaly scoring. In doing so, BUSTER makes VLM-guided anomaly detection without human annotations more practical for continuous surveillance, where both efficiency and selective temporal refinement are essential.
We summarize our contributions as follows:

- We introduce BUSTER, a distillation framework without human annotations that learns a lightweight student for scout-and-burst inference, allocating dense temporal inspection only to score-triggered candidates instead of processing the full stream at a fixed rate.
- We show that VLM-derived pseudo supervision can train a lightweight recurrent student not only for anomaly scoring but also for content-aware temporal budget allocation, and we analyze this design through teacher and student variant studies.
- We demonstrate strong efficiency and accuracy in long surveillance streams, achieving 496.7 FPS corresponding to 16.6× real-time processing at 30 FPS while attaining state-of-the-art performance among unsupervised methods, and we further report cross-dataset evaluation under domain shift.

2 Related Work

From Video Anomaly Detection to Video Anomaly Understanding. Recent research in video anomaly analysis has moved beyond anomaly scoring toward more semantic and interpretable understanding of abnormal events. Early vision-language based VAD methods such as CLIP-TSA [9] and VadCLIP [25] leverage vision-language priors to improve anomaly discrimination, while more recent works including LAVAD [32], VERA [29], Holmes-VAU [34], and AnomalyRuler [27] further extend this direction toward anomaly explanation and reasoning by incorporating large vision-language or multimodal foundation models [2, 4, 7, 10, 15, 35]. These methods demonstrate that heavy foundation models can provide richer semantic cues for understanding complex abnormal events, but they also expose a practical limitation: long surveillance videos remain difficult to process densely due to the substantial cost of frame-wise reasoning. As a result, most existing methods either operate on a small set of sampled frames or incur high latency when temporal coverage is increased. Holmes-VAU [34] improves efficiency by reallocating frame density according to anomaly scores from a lightweight scorer, yet its sampling policy is still inherited from generic anomaly scoring rather than learned as a controller for temporal budget allocation itself. In contrast, our method learns the lightweight controller directly from VLM-derived pseudo supervision, enabling adaptive temporal allocation without relying on human annotations.
Efficient Temporal Sampling for Long Surveillance Streams. Efficient video sampling

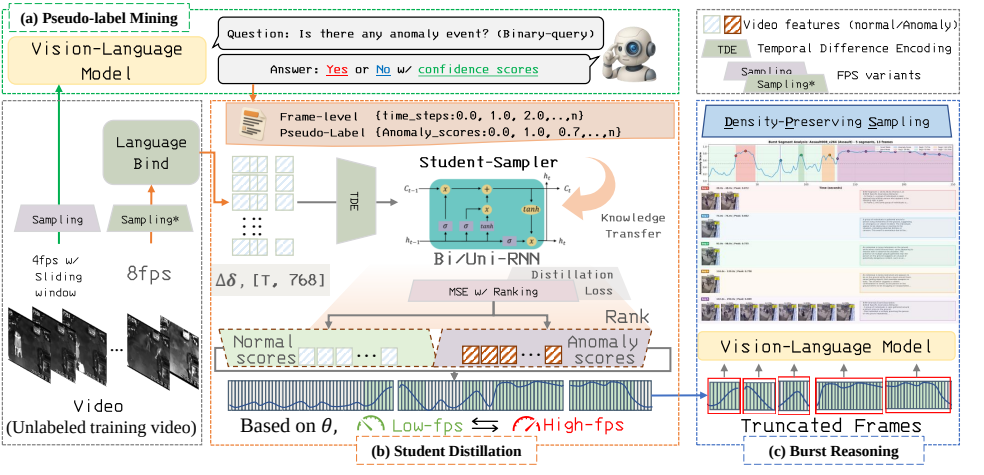


Figure 2: BUSTER distills confidence-based pseudo-labels from a VLM teacher (a) into a lightweight student that monitors the stream in *Scout* mode and switches to *Burst* sampling around anomaly candidates (b), then invokes the VLM only on the resulting burst clips to generate interpretable explanations under a fixed frame budget (c).

has been widely studied as a way to handle long videos under limited compute or context budgets. A common approach is uniform sparse sampling, which reduces computational cost but often suffers from temporal aliasing because short-lived anomaly cues may be missed, resulting in incomplete temporal evidence for downstream reasoning. Another line of work explores query-driven sampling, where keyframes are selected according to their relevance to a text query. Such strategies are effective for VideoQA or retrieval-style settings, where iterative query-conditioned retrieval and repeated frame re-access are often used to gather missing evidence [5, 8, 11, 13, 14, 28, 30, 37]. However, these designs can introduce latency and system overhead that are undesirable for persistent surveillance monitoring [8, 13]. More importantly, in long surveillance streams the core challenge is not only to reduce the number of processed frames, but also to decide *when* denser temporal inspection should be triggered and *where* additional temporal budget should be spent, since fixed-rate sampling cannot adapt temporal allocation to evolving video content. Our Scout-and-Burst design addresses this setting directly by using a lightweight student to monitor the stream at low cost and to selectively expand computation only around suspicious intervals, making adaptive temporal refinement a learned part of the inference pipeline.

3 Method

3.1 Overall Framework

Figure 2 illustrates the overall framework of BUSTER, which consists of three stages: (a) pseudo-label mining, (b) student distillation, and (c) burst reasoning.

In stage (a), the input video is first preprocessed through two sampling streams for dif-

ferent roles. For the teacher VLM, we sample the video at 4 FPS with overlapping temporal windows, where each 2-second window contains 8 uniformly sampled frames and adjacent windows overlap with a 1-second stride. This dense sliding-window construction is used to provide near-continuous temporal coverage with minimal frame omission, so that the teacher can assign anomaly-presence confidence to local video segments under a sufficiently fine temporal resolution. The teacher is prompted with a binary anomaly query, and its normalized Yes/No token probabilities are converted into window-level anomaly confidence scores, which are then aligned into frame-level pseudo-labels. Separately, for LanguageBind feature extraction, we sample the video at 8 FPS as a preprocessing step. This rate is chosen as a practical trade-off between efficiency and temporal fidelity for anomaly analysis in 30 FPS videos. LanguageBind produces semantic video features, and Temporal Difference Encoding (TDE) is then applied along the temporal dimension by computing feature deltas with respect to the previous segment. Although TDE does not explicitly model motion cues such as optical flow, it captures semantic variation across neighboring segments, thereby providing temporal sensitivity without expensive motion estimation and offering an important cue for anomaly score refinement.

In stage (b), the student sampler is implemented as a lightweight recurrent model based on an LSTM. Our default design uses a bidirectional LSTM to exploit future context within the video sequence, while unidirectional RNN or GRU variants can also be used when causal inference is preferred for real-time deployment. The student is trained with the teacher-derived pseudo-labels using an MSE distillation loss together with a ranking objective that enlarges the score gap between normal and anomalous segments. In this way, the student learns not only to imitate the teacher's anomaly scores, but also to refine them using temporal semantic variation from TDE. At inference time, the resulting student score serves as the control signal for scout-to-burst switching: normal segments remain in low-FPS Scout mode, whereas anomalous segments are revisited in high-FPS Burst mode. By allocating different sampling rates to different segments within the same video, BUSTER reduces unnecessary computation while preserving denser temporal evidence around suspicious events.

In stage (c), only the burst-selected anomalous segments are truncated and forwarded to the reasoning VLM. Before reasoning, these segments are further processed by Density-Preserving Sampling (DPS), which adjusts the number of reasoning frames according to the segment length under a fixed budget. Longer anomalous segments therefore receive more sampled evidence within the same budget, while shorter segments are processed more compactly. This enables dynamic reasoning over burst intervals of different durations and helps preserve temporal context even for relatively long anomalous events, while substantially reducing the latency of applying VLM reasoning to the full video.

3.2 Problem Definition

We study label-free video anomaly detection, where no ground-truth annotations are used for training. Given an untrimmed video $V = \{I_t\}_{t=1}^{T}$, our goal is to predict frame-level anomaly scores $\mathbf{s} = \{s_t\}_{t=1}^{T}$ with $s_t \in [0,1]$, where higher scores indicate anomalous frames. In practice, the student outputs $\hat{\mathbf{y}}^{S} = \{\hat{y}_t^{S}\}_{t=1}^{T'}$ on the feature timeline, which is mapped to the frame timeline to obtain $\mathbf{s} = \{s_t\}_{t=1}^{T}$ for evaluation.

3.3 Teacher Pseudo-Label Mining

Given an untrimmed training video $V = \{I_t\}_{t=1}^{T}$, we obtain window-level anomaly confidences from a VLM teacher and align them into frame-level pseudo-labels for student training. We scan the video with overlapping temporal windows of length $\tau = 2$ seconds and stride $\Delta = 1$ second. For each window w_k centered at time c_k, we uniformly sample $n = 8$ frames as input to the teacher. To compute an anomaly confidence for each window, we prompt the teacher with a binary anomaly-presence query, *Does any anomaly event occur in this video? Answer Yes or No.*, and extract the logits of the final response token. We intentionally use a binary prompt rather than asking for a specific anomaly category. In surveillance videos, anomaly types are often open-ended and visually diverse, and prompting the model to predict a predefined event label can overemphasize its category prior while underutilizing the actual visual context. By reducing the decision to anomaly presence, we instead seek a teacher signal that reflects contextual abnormality more directly, without forcing the model into a potentially oversimplified event taxonomy.

Let $\mathbf{z}_k \in \mathbb{R}^{|\mathcal{V}|}$ be the teacher logits over the vocabulary $\mathcal{V}$ for window w_k, and $\mathbf{p}_k = \mathrm{softmax}(\mathbf{z}_k)$ be the corresponding probability distribution. We define token sets $\mathcal{Y}$ and $\mathcal{N}$ that include common variants of "Yes" and "No" (e.g., leading spaces and different cases), and compute

$$p_k^{\mathrm{yes}} = \sum_{v \in \mathcal{Y}} \mathbf{p}_k[v], \quad p_k^{\mathrm{no}} = \sum_{v \in \mathcal{N}} \mathbf{p}_k[v]. \tag{1}$$

We then obtain the window-level anomaly score via binary normalization:

$$y_k^{\mathrm{T}} = \frac{p_k^{\mathrm{yes}}}{p_k^{\mathrm{yes}} + p_k^{\mathrm{no}} + \varepsilon}, \tag{2}$$

where ε is a small constant for numerical stability. This yields a continuous confidence $y_k^{\mathrm{T}} \in [0,1]$ without relying on text generation or parsing. Finally, the sequence $\{(c_k, y_k^{\mathrm{T}})\}_{k=1}^{K}$ is interpolated onto the student timeline to provide aligned frame-level pseudo-labels:

$$y_t^{\mathrm{T}} = \mathrm{Interp}\big(\{(c_k, y_k^{\mathrm{T}})\}_{k=1}^{K}\big)(t), \quad t = 1, \ldots, T', \tag{3}$$

where T' denotes the length of the student feature sequence. Additional explanations of the algorithms are provided in Supplementary A.2

3.4 Student Distillation with Scout-and-Burst Policy

Student input features. We extract compact video features at 8 FPS using a frozen LanguageBind encoder [58], denoted as $X = \{x_t\}_{t=1}^{T'}$ with $x_t \in \mathbb{R}^d$ and $d = 768$. To encode temporal changes, we compute a Temporal Difference Encoding (TDE) using first-order differences:

$$\delta x_t = x_t - x_{t-1}, \quad \delta x_1 = \mathbf{0}, \tag{4}$$

and form the augmented input

$$\tilde{x}_t = [x_t; \delta x_t] \in \mathbb{R}^{2d}. \tag{5}$$

Student architecture. The student projects $\tilde{x}_t$ into a hidden space and predicts a frame-level anomaly score via a lightweight recurrent encoder:

$$h_t = \mathrm{RNN}\big(\phi(\tilde{x}_t), h_{t-1}\big), \quad \hat{y}_t^{\mathrm{S}} = \sigma\big(g(h_t)\big) \tag{6}$$

where $\phi(\cdot)$ is an input projection, RNN$(\cdot)$ denotes an LSTM/GRU encoder, $g(\cdot)$ is an MLP scoring head, and $\sigma(\cdot)$ is the sigmoid function. This yields the student score sequence $\hat{\mathbf{y}}^{\mathrm{S}} = \{\hat{y}_t^{\mathrm{S}}\}_{t=1}^{T'}$, with $\hat{y}_t^{\mathrm{S}} \in [0,1]$.

Distillation objective. We train the student using aligned teacher pseudo-labels $\mathbf{y}^{\mathrm{T}} = \{y_t^{\mathrm{T}}\}_{t=1}^{T'}$ obtained by interpolating window-level teacher confidences. The overall objective combines a distillation loss, a ranking regularizer, and a temporal smoothness term:

$$\mathcal{L} = \mathcal{L}_{\mathrm{dist}} + \lambda_{\mathrm{rank}}\mathcal{L}_{\mathrm{rank}} + \lambda_{\mathrm{smooth}}\mathcal{L}_{\mathrm{smooth}}. \tag{7}$$

The distillation loss is defined as

$$\mathcal{L}_{\mathrm{dist}} = \frac{1}{|\Omega|}\sum_{t\in\Omega}\left(\hat{y}_t^{\mathrm{S}} - y_t^{\mathrm{T}}\right)^2, \tag{8}$$

where Ω denotes valid (non-padded) time indices. To preserve the relative ordering between high-confidence anomalous and normal evidence, we define $\mathcal{H} = \{t \in \Omega \mid y_t^{\mathrm{T}} > \tau_p\}$ and $\mathcal{N}_{\mathrm{neg}} = \{t \in \Omega \mid y_t^{\mathrm{T}} < \tau_n\}$, and sample pairs $\mathcal{P} = \mathcal{H} \times \mathcal{N}_{\mathrm{neg}}$. We apply a margin ranking loss:

$$\mathcal{L}_{\mathrm{rank}} = \frac{1}{|\mathcal{P}|}\sum_{(a,n)\in\mathcal{P}} \max\left(0, m - (\hat{y}_a^{\mathrm{S}} - \hat{y}_n^{\mathrm{S}})\right), \tag{9}$$

where $\hat{y}_a^{\mathrm{S}}$ and $\hat{y}_n^{\mathrm{S}}$ denote the student scores at indices a and n, respectively, and m is a margin.

Scout-and-burst inference. At inference time, we distinguish between an offline evaluation view and a deployment-time implementation. For offline evaluation with pre-extracted features, scout mode is represented as a 4 FPS sub-sampling of the same 8 FPS feature stream:

$$x_u^{\mathrm{scout}} = x_{2u}, \quad u = 1, \ldots, \lfloor T'/2 \rfloor, \tag{10}$$

and scout scores $\hat{y}_u^{\mathrm{scout}}$ are obtained using the same student model. If $\hat{y}_u^{\mathrm{scout}}$ exceeds a threshold θ, we trigger burst processing and re-evaluate the corresponding interval at 8 FPS to obtain dense student scores. In the actual deployment implementation, this corresponds to extracting LanguageBind features directly at 4 FPS during scout monitoring, and re-extracting features at 8 FPS only for burst-triggered windows. Additional explanations of the algorithms are provided in Supplementary A.1.

3.5 Segment-level VLM Reasoning

At inference time, the burst-region indices on the 8 FPS student timeline are first merged into a set of candidate segments $\mathcal{S} = \{[a_m, e_m]\}_{m=1}^{M}$, where $1 \le a_m \le e_m \le T'$ and M is capped by a fixed maximum. Let $\ell_m = e_m - a_m + 1$ denote the length of segment m.

Budget allocation. Given a global frame budget B, we allocate a per-segment sampling budget proportional to its duration:

$$q_m = \min\left(\ell_m,\ q_{\max},\ \max\left(1, \lfloor B \cdot \frac{\ell_m}{\sum_{j=1}^{M}\ell_j} \rfloor\right)\right), \tag{11}$$

where $q_{\max}$ limits the maximum number of frames per segment.

Density-preserving sampling. Within each segment $[a_m, e_m]$, we define the normalized student-score density

$$p_t^{(m)} = \frac{\hat{y}_t^S}{\sum_{j=a_m}^{e_m}\hat{y}_j^S}, \qquad I_m \sim \mathrm{Sample}\left([a_m, e_m], q_m; p^{(m)}\right), \tag{12}$$

so that higher-confidence frames are sampled more densely. We additionally ensure inclusion of the peak-score time $t_m^\star = \arg\max_{t\in[a_m,e_m]} \hat{y}_t^S$:

$$I_m \leftarrow \text{ReplaceClosest}(I_m, t_m^\star). \tag{13}$$

The selected frames form a short sequential clip $\tilde{V}_m = \{I_t\}_{t\in I_m}$, which is fed to the VLM for segment-level reasoning.

4 Experiments

4.1 Datasets and Evaluation Metrics

We evaluate BUSTER on three widely used untrimmed video anomaly benchmarks: UCF-Crime [17], ShanghaiTech [12], and XD-Violence [24]. Across all datasets, we follow a training protocol without human-provided temporal annotations or video-level anomaly/normal labels, and use the available test-time temporal annotations only for evaluation.

UCF-Crime. UCF-Crime is a large-scale real-world surveillance benchmark containing 1,900 untrimmed videos with a total duration of 128 hours and 13 anomaly categories. The standard split contains 810 anomalous and 800 normal training videos, and 140 anomalous and 150 normal test videos.

ShanghaiTech. ShanghaiTech Campus contains 13 scenes with diverse lighting conditions and viewpoints. It includes 330 training videos and 107 test videos, with over 270k training frames and 130 abnormal events. We use its frame-level annotations only for evaluation.

XD-Violence. XD-Violence is a large-scale long-form violence benchmark containing 4,754 untrimmed videos with a total duration of 217 hours. It covers 6 violent-event categories and includes audio signals, with videos collected from movies and in-the-wild sources. We follow the standard split of 3,954 training videos and 800 test videos.

Evaluation metrics. Following standard VAD protocols, we report frame-level ROC-AUC and frame-level AP. ROC-AUC measures how well the predicted anomaly scores separate normal and abnormal frames across thresholds, while AP summarizes the precision-recall trade-off and is particularly informative under class imbalance. In addition, for ablation and thresholded analysis, we report the false alarm rate (FAR) to measure how often alarms are raised on normal frames under a fixed decision threshold. Additional implementation details and hyperparameter settings are provided in the Supplementary A.5 and B.6.

4.2 Benchmark Results and Practical Efficiency

Comparison on standard benchmarks. Table 1 compares BUSTER with prior VAD methods across different supervision regimes. Among unsupervised methods, BUSTER achieves the best results on UCF-Crime in both Offline and Online modes, and also sets a new best Offline result on ShanghaiTech. On XD-Violence, BUSTER delivers strong competitive AUC and substantially improves AP over prior unsupervised methods, showing that the proposed framework remains effective on long-form violent-event videos as well. Notably, these gains are obtained while maintaining interpretability and supporting both Offline and causal Online inference.

Efficiency of adaptive sampling. To improve transparency, Table 2 reports analytical FPS

Table 1: State-of-the-art comparisons on VAD benchmarks: UCF-Crime, XD-Violence, and ShanghaiTech. Methods are grouped by supervision level. For BUSTER, improvements/declines are computed relative to the strongest prior unsupervised baseline for each dataset and metric, and are highlighted in green/red.

Methods	Interpretable	Mode	UCF-Crime	XD-Violence		ShanghaiTech
			AUC (%) ↑	AUC (%) ↑	AP (%) ↑	AUC (%) ↑
Weakly-Supervised w/o VLM						
Sultani et al. [12] (CVPR'18)	✗	Offline	77.92	-	-	-
RTFM [20] (ICCV'21)	✗	Offline	84.03	-	77.81	97.21
GCL [31] (CVPR'22)	✗	Offline	79.84	-	-	79.62
S3R [23] (ECCV'22)	✗	Offline	85.99	-	80.26	79.62
MGFN [6] (AAAI'23)	✗	Offline	86.98	-	80.11	93.90
UR-DMU [36] (AAAI'23)	✗	Offline	86.97	94.02	81.66	85.02
TSRN [8] (TIP'25)	✗	Offline	87.90	-	-	98.20
Weakly-Supervised w/ VLM						
CLIP-TSA [9] (ICIP'23)	✗	Offline	87.58	-	82.17	-
VadCLIP [25] (AAAI'24)	✗	Offline	88.02	-	84.51	88.48
Holmes-VAD [33] (arXiv'24)	✓	Offline	84.61	-	84.96	-
Holmes-VAU [34] (CVPR'25)	✓	Offline	88.96	-	87.68	-
VERA [29] (CVPR'25)	✓	Offline	86.55	88.26	70.54	-
Zero-shot						
LAVAD [32] (CVPR'24)	✓	Offline	80.28	85.36	62.01	44.34
AnomalyRuler [27] (ECCV'24)	✓	Offline	-	-	-	85.20
Unsupervised						
GCL [31] (CVPR'22)	✗	Offline	74.20	-	-	-
Tur et al. [21] (ICIP'23)	✗	Offline	65.22	-	-	76.10
DyAnNet [19] (WACV'23)	✗	Offline	79.76	-	-	-
RareAnom [13] (PR'23)	✗	Offline	79.76	-	68.33	-
BUSTER (Ours)	✓	Offline	**86.30** (+6.54)	**93.93**	**80.48** (+12.15)	**78.30** (+2.20)
		Online	**85.50** (+5.74)	**93.80**	**80.35** (+12.02)	**75.90** (-0.20)

Table 2: Estimated efficiency of adaptive sampling. FPS and RTF are computed from measured module latencies.

Method	AUC (%)	Data Usage	LB feat.	VLM	Burst regime	FPS	RTF
Uniform Sampling							
Total 8 frames	52.32	8 frames†	N/A	N/A	fixed count	N/A	N/A
Total 16 frames	55.58	16 frames†	N/A	N/A	fixed count	N/A	N/A
Total 32 frames	63.39	32 frames†	N/A	N/A	fixed count	N/A	N/A
Fixed Sampling							
Full-4fps	84.95	50.0%	on-the-fly	no	dense 4fps	528.2	17.6×
Full-8fps	86.30	100.0%	on-the-fly	no	dense 8fps	264.1	8.8×
Adaptive Sampling w/ Detection-only							
BUSTER Standard-load (dataset)	**86.30**	**53.2%**	on-the-fly	no	1.4% burst, 98.6% scout	**496.7**	**16.6**×
Adaptive Sampling w/ VLM							
BUSTER Low-load (90% normal)	86.30	59.6%	on-the-fly	yes	10% burst	351.1	11.7×
BUSTER High-load (100% anomaly)	86.30	100.0%	on-the-fly	yes	100% burst	109.0	3.6×

† Uniform sampling uses a fixed frame count regardless of video duration, so FPS/RTF is not directly comparable.

/RTF estimates from measured module runtimes, explicitly distinguishing cached versus on-the-fly LanguageBind features and the inclusion of VLM reasoning under different burst loads. Specifically, we measure LanguageBind feature extraction at 8.8 clips/sec on a single RTX A6000 GPU (batch size 16), the student LSTM at 4.1 ms for a sequence of length $T=100$, and the teacher VLM module at approximately 900 ms per window, i.e., about 1.1 clips/sec. Using these measured module latencies, we estimate FPS and real-time factor

Table 3: Comparison of computational efficiency and performance.

Method	Architecture	Supervision	FPS	Params	FLOPs	AUC (%)
UR-DMU	3D CNN	Weakly	316.1	6.49M	4.1G	**86.97**
LAVAD	CLIP ViT-B/16	Zero-shot	504.0	300M†	50G†	80.28
DyAnNet	3D CNN+Flow	Unsupervised	30†	30M†	20G†	79.76
Tur et al.	Linear	Unsupervised	1,332	2.37M	0.47G	65.22
BUSTER (Ours)	LSTM (Student)	Unsupervised	**496.7**	**3.19M**	**0.32G**	86.30

† Not explicitly reported in the original paper; estimated from the published architecture/backbone.

Table 4: Ablation study on Teacher-Student model variants.

Method	UCF-Crime		ShanghaiTech	
	AUC (%)	FAR@0.1 (%)	AUC (%)	FAR@0.1 (%)
Teacher: Qwen2.5-VL-32B [■]				
Teacher (VLM-only)	82.60	15.67	68.57	12.0
Student (LSTM)	86.30	**0.86**	**78.32**	**0.47**
Student (GRU)	**86.68**	1.45	75.90	0.72
Teacher: Qwen2.5-VL-72B [■]				
Teacher (VLM-only)	84.38	11.78	67.89	1.17
Student (LSTM)	86.11	0.52	72.40	< 0.01
Student (GRU)	**87.15**	**0.44**	**82.57**	$<$ **0.01**

(RTF) under different burst regimes. Here, *RTF* denotes the real-time factor with respect to 30 FPS video, and *High-Load* refers to an extreme setting with frequent anomaly bursts. Compared to dense Fixed-8fps, BUSTER preserves the same AUC (86.30) while reducing data usage to 53.2% in detection-only mode, indicating that adaptive sampling can maintain accuracy with substantially lower processing cost. When on-demand VLM is enabled for explanation, the realistic setting (90% normal) remains efficient, whereas the high-load setting quantifies the worst case in which frequent bursts push the system close to dense processing and expose the added cost of VLM reasoning.

Efficiency-performance trade-off. Table 3 situates BUSTER within the efficiency and performance landscape by comparing throughput, parameter count, FLOPs, and detection accuracy against representative prior methods. The comparison shows that BUSTER is not only effective under the proposed adaptive sampling policy, but also highly compact and computationally efficient at the model level. In particular, BUSTER achieves 496.7 FPS with only 3.19M parameters and 0.32G FLOPs, while maintaining competitive detection accuracy relative to substantially larger or more expensive baselines. Compared with unsupervised alternatives such as DyAnNet [19] and Tur et al. [21], BUSTER attains markedly stronger AUC with a similarly lightweight or lower-compute footprint, and it remains close to the weakly-supervised UR-DMU [36] despite operating with a smaller model and lower computational cost. These results indicate that the proposed student controller achieves a favorable trade-off between efficiency and detection quality, making it well suited to long-video anomaly understanding under practical compute constraints. Additional quantitative evidence is provided in the Supplementary B.

4.3 Ablations and Generalization

Teacher-student design choices. Table 4 studies the effects of teacher scale and student backbone under the same training protocol without human annotations. To minimize confounding factors, we vary teacher scale within a single VLM family, Qwen2.5-VL, from 32B to 72B, thereby isolating the effect of teacher-side pseudo-label quality. The results show that stronger teacher supervision improves downstream detection only when the stu-

Table 5: Comprehensive ablation study on UCF-Crime.

Model Variant	Input & Arch.		Objectives		Backbone	Performance Metrics		
	Δ Feat.	Bi-dir.	$\mathcal{L}_{Rank}$	$\mathcal{L}_{Smooth}$	RNN	AUC (%)	FAR@0.1 (%) ↓	Noise (N_{mean}) ↓
Baseline (Ours)								
BUSTER (Ours)	✓	✓	✓	✓	LSTM	**86.30**	0.89	**0.0033**
Architecture Ablations								
w/o Delta Features	✗	✓	✓	✓	LSTM	84.51 (-1.79)	1.52	0.0087
Streaming (Uni-dir)	✓	✗	✓	✓	LSTM	85.54 (-0.76)	3.53	0.0131
GRU Variant	✓	✗	✓	✓	GRU	86.56 (+0.26)	**0.84**	**0.0030**
Loss Ablations								
w/o Ranking Loss	✓	✓	✗	✓	LSTM	85.46 (-0.84)	2.31	0.0301
w/o Smoothness	✓	✓	✓	✗	LSTM	85.90 (-0.40)	3.14	0.0160

Table 6: Cross-dataset validation using target-domain gap.

Direction	LSTM			GRU		
	Target reference	Transferred	Target Gap	Target reference	Transferred	Target Gap
UCF → SHT	78.30	68.98	9.32	75.90	78.96	**-3.06**
SHT → UCF	86.30	74.94	11.36	86.68	75.58	11.10
UCF → XDV	93.93	93.05	0.88	93.80	92.96	0.84
XDV → UCF	86.30	86.16	**0.14**	86.68	84.69	1.99

Table 7: Effect of Scout/Burst temporal resolutions on the efficiency-accuracy trade-off.

Config	AUC (%)	FAR@0.1 (%) ↓	Data Usage	FPS	RTF
Burst = 4 fps					
2/4	84.92	1.09	26.7%	988.3	32.9×
4/4	84.95	1.10	50.6%	521.9	17.4×
Burst = 8 fps					
2/8	86.15	**0.75**	**28.6%**	**922.4**	**30.7**×
4/8 (Ours)	**86.30**	0.86	53.2%	496.7	16.6×
8/8	**86.30**	0.89	100.0%	264.1	8.8×

dent has sufficient capacity to absorb it. While the LSTM student benefits only marginally from the 72B teacher, replacing it with a GRU yields the best overall performance. Here, *FAR@0.1* measures the false alarm rate on *normal* frames at a strict threshold of 0.1, providing a conservative test of robustness to spurious alarms. Overall, the framework remains flexible to both teacher scaling and student substitution, supporting its *plug-and-play* design.

Component-wise contributions. Table 5 quantifies how each design choice affects both accuracy and alarm reliability on UCF-Crime. Temporal-delta inputs are critical for capturing short-term changes: removing Δ features drops AUC by 1.79 points and increases FAR@0.1, indicating more frequent false alarms under the strict threshold. Likewise, enforcing causal streaming notably amplifies spurious activations (FAR@0.1 ↑ to 3.53) and raises N_{mean}, where N_{mean} denotes the mean predicted anomaly score on normal frames. In the loss design, removing $\mathcal{L}_{Rank}$ or $\mathcal{L}_{Smooth}$ substantially increases FAR and N_{mean}, confirming that ranking sharpens anomalous-vs-normal separation while temporal smoothness suppresses noisy peaks during pseudo-label distillation.

Cross-dataset robustness. Table 6 reports cross-dataset transfer across UCF-Crime, ShanghaiTech, and XD-Violence using the *target-domain gap*, i.e., the difference between target in-domain performance and transferred performance. This evaluation is important in our setting because the framework is trained without human annotations and instead relies on VLM-derived pseudo supervision for a lightweight switching controller. The results show that transfer remains domain-dependent: UCF–ShanghaiTech exhibits much larger gaps, likely due to differences in camera motion, scene statistics, visual quality, and anomaly composition, whereas UCF–XD-Violence shows only small target-domain gaps. The GRU variant is

more robust than the LSTM on the more difficult UCF–ShanghaiTech transfer, suggesting that cross-domain robustness depends not only on teacher-side supervision, but also on student design.

Design choice of Scout/Burst temporal resolutions. We further study Scout/Burst frame rates to justify the adopted 4/8 design as a practical operating point. Table 7 shows that detection quality is primarily determined by the Burst rate: all Burst=8fps settings remain close to dense 8/8 performance, whereas Burst=4fps causes a clear drop in AUC. Within the Burst=8fps group, reducing the Scout rate from 4fps to 2fps improves efficiency further, but at the cost of a small reduction in AUC. We therefore adopt 4/8 as the most balanced setting, since it preserves the dense 8/8 AUC exactly while still reducing data usage by nearly half.

4.4 Visualization

Figure 3 presents qualitative examples from UCF-Crime `Assault008` and `Robbery008`, as well as from the XD-Violence sample `Casino.Royale.2006`. Each row shows the student anomaly score sequence $\hat{y}_t^S$, burst-triggered candidate segments, density-preserving sampled frames, and segment-level descriptions generated by the reasoning VLM, Qwen2.5-VL-32B. Under the realistic low-load setting with on-demand VLM reasoning, the full pipeline operates at 351.1 FPS, as reported in Table 2. The examples show that the proposed sampling strategy preserves enough temporal evidence for the VLM to produce contextually grounded descriptions, capturing salient objects, environment, interactions, and anomaly cues. This is particularly evident not only in relatively direct violent events such as `Assault008`, but also in cases that require finer temporal reasoning, such as `Robbery008`, where object-person interactions unfold over time, and the XD-Violence movie clip, where the model captures a richer narrative context beyond a single anomaly label. These examples further suggest that effective temporal sampling helps reduce failure modes caused by biased decisions or poorly selected evidence, thereby mitigating hallucination in downstream VLM reasoning. Overall, the generated descriptions align well with the observed event dynamics within the burst regions, suggesting that scout-and-burst inference can support both efficient detection and semantically meaningful explanation. Additional qualitative examples, including normal samples and further anomaly classes, are provided in the Supplementary C.

5 Discussion

Table 8 compares the scoring module and the refinement policy under matched conditions. All rows use the same LanguageBind 8fps features, pseudo supervision, smoothing, interpolation, and approximately matched total usage, thereby isolating anomaly scoring from refinement allocation. The near-identical AUC and AP across refinement policies are expected, since both policies share the same 4fps scout pass, the same scorer, and the same overall budget. In this setting, global frame-level detection metrics are largely determined by the shared scoring backbone, while the refinement policy mainly affects where the additional 8fps evidence is placed. Accordingly, the clearest differences appear in **Video Hit Rate**, **Event Recall**, and **Refinement Precision**. The large gain in Refinement Precision does not necessarily imply an equally large gain in AUC or AP, because these quantities measure different aspects of the pipeline: Refinement Precision evaluates the quality of the small extra refinement budget, whereas AUC and AP are computed over the full frame sequence. Under this view, the proposed Scout-and-Burst policy consistently outperforms the

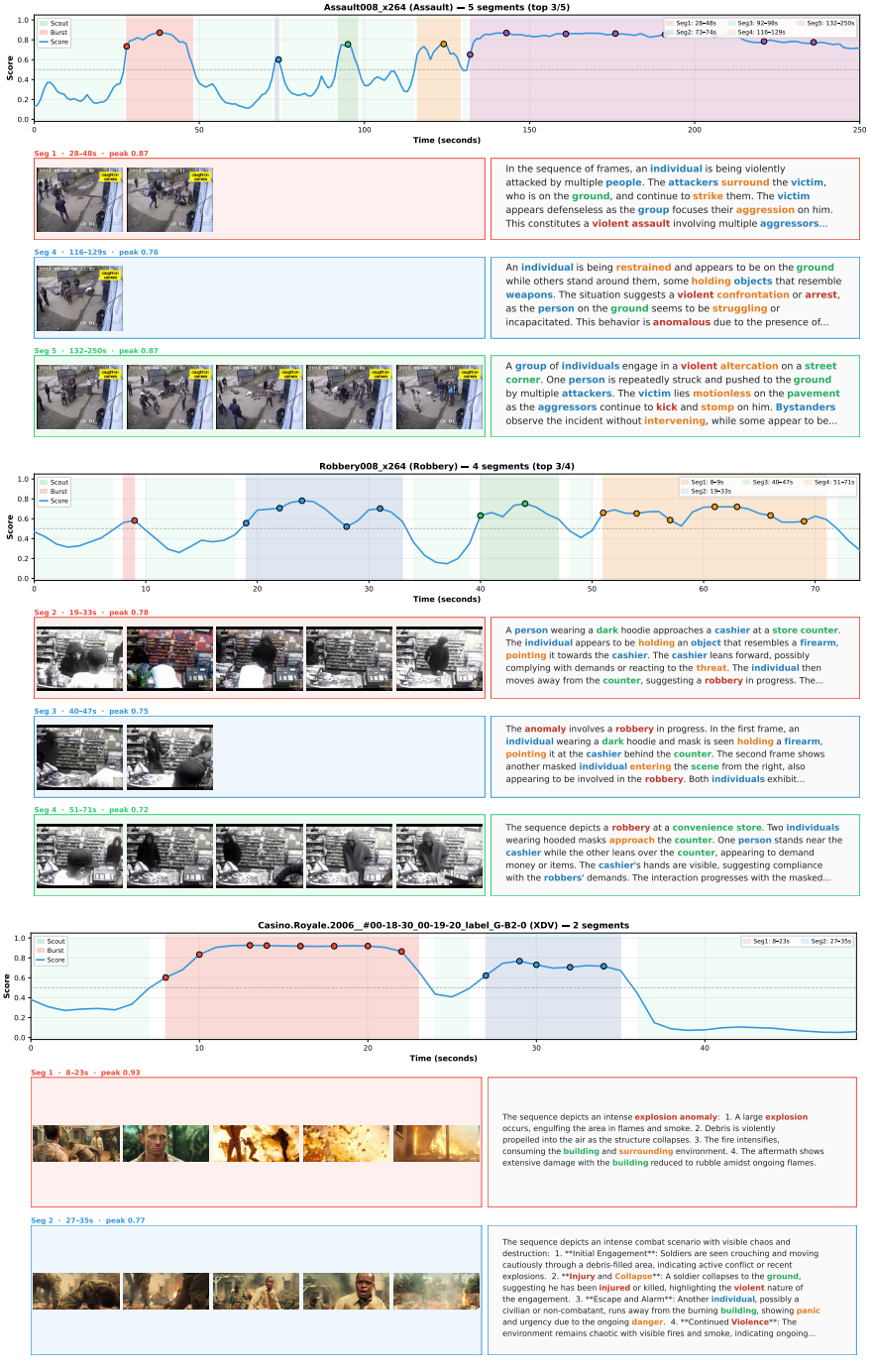


Figure 3: Qualitative scout-and-burst examples with on-demand VLM reasoning. Each row shows the score curve, burst-triggered segments, sampled frames, and segment-level descriptions. Color-coded words indicate objects, environment, interactions, and anomalies.

Table 8: **Controlled comparison of scoring modules and refinement policies under matched conditions on UCF-Crime.** All rows use the same LanguageBind 8fps features, pseudo supervision, smoothing, and interpolation. *Holmes-style* denotes the score-driven redistribution rule adapted to the shared pipeline.

Scoring Module	Refinement Policy	Detection Metrics			Refinement Metrics			
		AUC (%)	AP (%)	FAR@0.1 (%)	Total Usage	Video Hit Rate (%)	Event Recall (%)	Refinement Precision (%)
UR-DMU -backbone	Holmes-style	86.10	37.32	16.07	52.9%	17.1	15.7	5.0
	Scout-and-Burst	**86.11**	**37.46**	16.08	53.1%	**22.1**	**22.1**	**52.3**
BUSTER recurrent	Holmes-style	85.93	37.64	14.07	52.9%	15.7	13.9	4.7
	Scout-and-Burst	**85.94**	**37.75**	14.07	53.3%	**25.7**	**25.4**	**55.9**

Holmes-style score-redistribution policy for both scoring modules, showing that the main benefit of BUSTER lies not in altering coarse anomaly scores, but in allocating the same refinement budget much more precisely to anomaly-relevant regions.

6 Limitations

Although BUSTER avoids human annotations by distilling a strong teacher VLM, it inevitably inherits biases from the teacher's pretrained knowledge and prompt prior. Our binary anomaly-presence prompt was chosen to avoid overcommitting to dataset-specific event labels and to remain applicable to open-set, in-the-wild surveillance settings, where anomaly categories, appearance, and contextual cues vary substantially across domains. However, this design also introduces limitations: the teacher may make conservative judgments, miss subtle anomalies, or rely on prior or posterior context rather than the decisive event itself. These failure modes are particularly visible when anomaly evidence is visually small, relational, weakly localized, or culturally/contextually ambiguous, such as tiny objects, complex human interactions, consequences mistaken for causes, or domain-specific notions of abnormality. Teacher-side bias, calibration sensitivity, and domain shift therefore remain important concerns for deployment and future work.

7 Conclusion

In this work, we presented BUSTER, a label-free VLM-guided framework for efficient video anomaly detection that addresses the practical bottleneck of applying heavy reasoning models to long surveillance streams. By distilling a VLM teacher into a lightweight student and using a Scout-and-Burst policy, BUSTER preserves strong detection performance while substantially reducing processed frames, and avoids the irreversible information loss of hard early pruning by revisiting suspicious intervals at higher temporal resolution. This explicit sampling design makes VLM reasoning both more efficient and more reliable, by concentrating computation on temporally relevant evidence and reducing hallucination from poorly selected context. Beyond single-stream evaluation, such low-cost Scout monitoring and selective Burst verification are especially well suited for multi-camera deployments, where practical scalability depends on allocating compute only when and where it is needed.

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